MULTIMEDIA UNIVERSITY

STUDENT ID NO

MULTIMEDIA UNIVERSITY

FINAL EXAMINATION

TRIMESTER MARCH/APRIL 2025 (TERM ID 2510)

CCP4213 – DATA STRUCTURE AND ALGORITHMS

(All sections / Groups)

17 JULY 2025

9.00 a.m - 11.00 a.m
(2 Hours)

INSTRUCTIONS TO STUDENTS

1. This question paper consists of five (5) pages, excluding the cover page.
2. There are five (5) questions. Answer ALL questions. The total is 50 marks.
3. Write your answers in the Answer Booklet provided.

QUESTION 1 [10 Marks]

a) Given the following code segment in Figure 1, trace the output : (2 marks)

```
int a[]={2,20,14,6};
int n = 18;
int &v = n;
int *p1, *p2;

p1 = a;
p2 = &n;
*(p1+3) = *p1* (*p2+1);
*p2 = *(p1+2) * 3;
p1[1] = a[0] * 4;
*p2 = v / 3;

for (int i = 3; i>=0; i--){
     cout<<*(p1+i) - v<<" ";
}
```

Figure 1

b) Transform the following *infix* notation to *postfix* notation. (1 mark)

$$(A+(B-D))*C/A$$

c) Given that A = 2, B = 4, C = 6, D = 1, based on your answer in b), evaluate the *postfix* notation using stack. (2 marks)

d) Figure 2 shows a stack with an array implementation named *myStack,* which stores integer values. A variable named *topStack* is used to keep track of the top element in the stack. Complete the code segment below to add a new item to the stack. (2 marks)

myStack

[0]	[1]	[2]	[3]
2	4	3	

topStack 2

```
int push(----(I)----)
{
    if (!full()){
        ------(II)-------
        ------(III)------
    }
    else{
        ------(IV)------
    }
}
```

Figure 2

Continued .....

e) Refer to Figure 3; *Q* is a *queue* that can store four integers, while *S* is a *stack* that can also store four integers. *A* and *B* are variables that store values 4 and 8, respectively.

Trace and draw the contents of *S*, *Q*, *A* and *B* after each of the following operations.

The drawing must clearly show and use the updated content of *S*, *Q*, *A* and *B* in each of the subsequent operations. (3 marks)

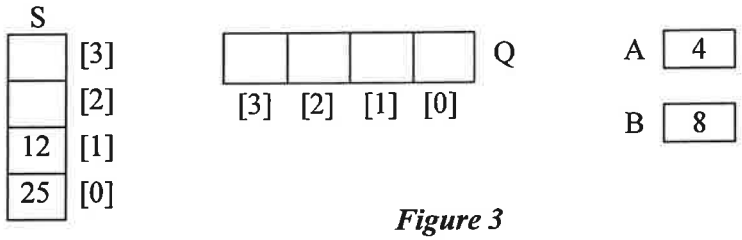

Figure 3

i.	`Q.enqueue(A + B);`
ii.	`if(S.pop()> Q.enqueue(B))` `  S.push(Q.dequeue());`

iii.	`if(Q.dequeue()< S.pop())`
	`  Q.enqueue(A)`
	`else`
	`  B = S.pop()`

QUESTION 2 [10 Marks]

a) Table 1 represents an array implementation of a linked list.

Table 1

Index	Data	Link
0	23	
1	45	
2	12	
3	65	

Taking 0 as the start of the list and 99 as a dummy representing the end of the list, fill in the link for all the elements in the table to maintain a list of the following order. (2 marks)

23 65 45 12

Continued .....

b) Given the following structure and variable declarations in Figure 4, complete the code segment in Figure 5 that displays data from a linked list. You can only use variables and pointers declared in Figure 4. (2 marks)

```
struct node{
      int data;
      node *next;
};
node *head, *temp;
```

Figure 4

```
------(I)------

while (-----(II)-----)
{
      cout<<-----(III)-----<<" ";

      -----(IV)------
}
```

Figure 5

c) Given the following structure and variable declarations in Figure 6. and a linked list diagram in Figure 7, answer all the following questions. You can only use variables and pointers declared in Figure 6. The pointer *head* must always point to the first node in the linked list.

```
struct node{
      int data;
      node *next;
};

node *head, *temp, *prev;
```

Figure 6

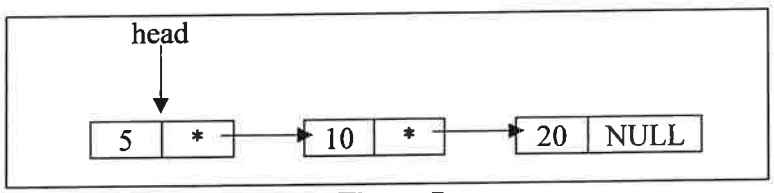

Figure 7

i. Write the code segment to add a new node with data 15 between the nodes with data 10 and 20. (3 marks)

ii. Based on the updated diagram in c (i), write the code segment to delete the node with data 10. The *temp* pointer is currently pointing to the *head* node. (3 marks)

Continued .....

QUESTION 3 [10 Marks]

a) Using the *binary search algorithm*, demonstrate step by step how to find the data with the value 20 in the given array in Figure 8. Write down the values of *first*, *last*, and *middle* at each step. (3 marks)

71	62	54	43	35	20
[0]	[1]	[2]	[3]	[4]	[5]

Figure 8

b) Write a hashing program that prompts an employee to enter his/her employed number of 7-digit long (key value). The system must calculate the hash value by truncating the first two digits and the last three digits. Then, display the hash value. (2 marks)

c) List two methods of resolving collisions by *open addressing*. (1 mark)

d) Assume a hash table with 9 locations and the hashing function $h(x) = x \% 9$. Show the index for each key and the result of the hash table when the following integers are inserted in the order given. Use *Chaining* to resolve the collision. (4 marks)

6, 13, 56, 22, 33

QUESTION 4 [10 Marks]

a) Figure 9 shows an array of five elements. Show the steps to sort the elements in ascending order by using *Merge Sort*. (4 marks)

25	6	98	43	17

Figure 9

b) Based on the binary tree in Figure 10, answer all the following questions:

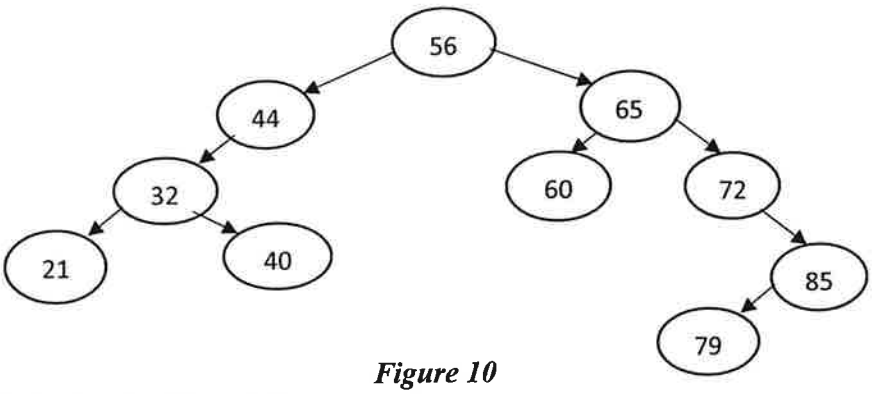

Figure 10

i. List the leaf node of the tree? (1 mark)
ii. Write the PostOrder traversal. (1 mark)
iii. Redraw the tree when node with data 70 is inserted and data 85 is deleted. (2 marks)

c) Draw a binary tree based on the following algebraic expression: (2 marks)

((H + J) / K) + G * H

Continued .....

QUESTION 5 [10 Marks]

a) Given the graph in Figure 11, answer all the following questions :

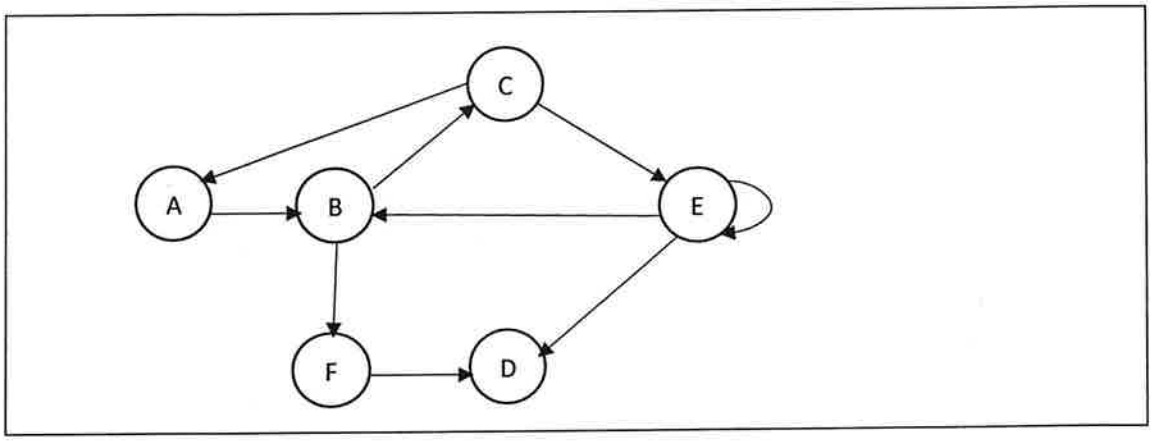

Figure 11

i. What is the type of graph? (1 mark)
ii. List all adjacent node of node B. (2 marks)
iii. Represent the graph using an adjacency matrix. (3 marks)
iv. List TWO simple cycles that you can identify in the graph. (2 marks)

b) Referring to Figure 12, traverse the graph using *breadth first search* and *depth first search* starting at vertex C. (2 marks)

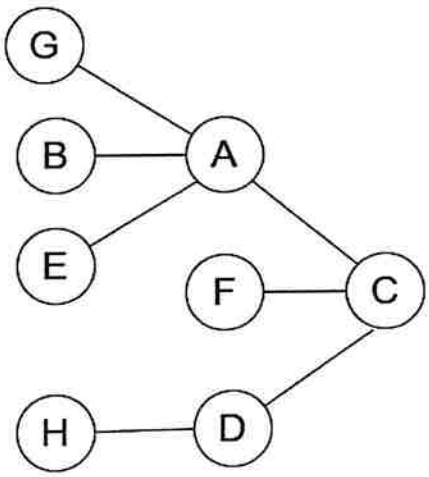

Figure 12

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